higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 2

(6021032)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat and in proportion
5. Start each question on a NEW page.
6. Write neatly and legibly.

NCV

QUESTION 1

1.1 Choose an/a item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–K) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Causes a wheel to wobble from side to side	A	ignition system
1.1.2	A pre-check before using lifting equipment	B	radians
1.1.3	Used to install a bearing on a shaft	C	sulphuric acid
1.1.4	Creates the spark for combustion	D	air pollution
1.1.5	An open can of thinners	E	weight capacity
1.1.6	Unit of angular measurement	F	viscosity
1.1.7	It can be harmful for your skin	G	radiator cap
1.1.8	The ability of oil to flow	H	hydraulic press
1.1.9	It controls the cooling system pressure	I	hone
1.1.10	Used to clean the cylinder bore	J	dynamic imbalance
		K	level

(10 × 1) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 It is the duty of the employer to ensure that staff is trained about health and safety issues.

1.2.2 The distributor cap is part of the primary circuit of the ignition system.

1.2.3 '10W40' indicates that the oil can only be used in cold temperatures.

1.2.4 Anti-friction bearings that are not sealed should only be fitted inside a housing that contains a lubricant.

1.2.5 Changing the wheels from steel rims to magnesium rims will increase the un-sprung mass.

(5 × 1) (5)

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1.3 Complete the following sentences by using the words provided in the list below. Write only the word(s) next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

safety shoes; accident; handle; dressing; incident; safety signs

1.3.1 Display … to help prevent accidents or incidents.

1.3.2 Always wear … when working in a workshop.

1.3.3 An/A … is an unplanned event that occurs where there is damage to property.

1.3.4 Trim the grinder's stones neatly by using a … tool.

1.3.5 Do not tighten the … of the engineering vice with a hammer.

(5 × 1) (5)

1.4 Name any TWO substances that can cause fires that can occur in a motor workshop. (2)

1.5 The health and safety of employers and employees are protected by a set of regulations within the OHS Act.

1.5.1 Name TWO responsibilities of the employee mentioned in act. (2)

1.5.2 What does the acronym OHS represent? (1)

1.6 Identify the correct tool that is used to perform the following tasks.

1.6.1 To clamp the piston rings to install the piston.

1.6.2 To measure the crankshaft journal for wear.

1.6.3 It is used with a drilling machine to clean the cylinder bores.

1.6.4 To measure the oil clearance between the crankshaft bearings.

(4 × 1) (4)

1.7 1.7.1 Convert 1,32 meters to millimetres

1.7.2 Convert 5 100 grams to kilograms

(2 × 1) (2)

[31]

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QUESTION 2

2.1 Name FOUR steps you should follow to safely jack up and secure the front end of a vehicle when using a trolley jack.

NOTE: The work area and the vehicle have been prepared already. (4)

2.2 Explain what happens inside the two-stroke engine as the piston moves from top to bottom. (5)

2.3 Identify the parts listed in FIGURE 1 below. Only write the name of the part name next to the letters (A–C) in the ANSWER BOOK.

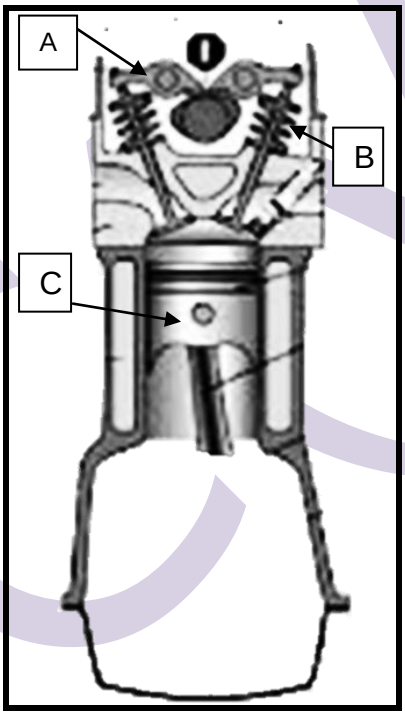


FIGURE 1 (3)

2.4 Explain the purpose of viscosity modifiers used in motor oils. (3)

2.5 Briefly explain how the heat that is generated inside the engine is used to benefit the operation of the engine. (3)

2.6 Name the TWO components that prevent dirt particles from entering the lubrication system of an engine. (2)

2.7 Name TWO advantages of an electric fan over a mechanical engine cooling fan. (2)

2.8 Explain why anti-freeze must be added to the coolant of a motor vehicle. (2)

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QUESTION 3

3.1 When the un-sprung mass is decreased, it makes a difference to the quality of the driving experience.

3.1.1 Give TWO advantages of the decreased un-sprung mass.

3.1.2 State TWO methods how a lighter un-sprung mass can be achieved.

(2 × 2) (4)

3.2 Use the images of the bearings in FIGURE 2 below to answer the questions.

3.2.1 Name the anti-friction bearings. (2)

3.2.2 Give ONE advantage that bearing B has over bearing A. (1)

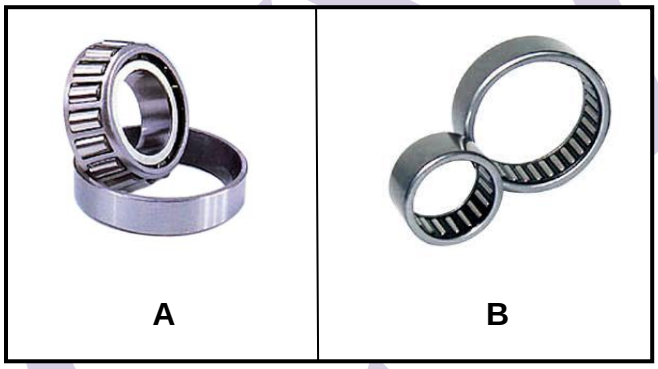


FIGURE 2

3.3 What tool is used to ensure that a wheel is properly secured when it is fitted to a car? (1)

3.4 Name the cause for each of the tyre wear patterns. Write your answer next to the letters (A–C).

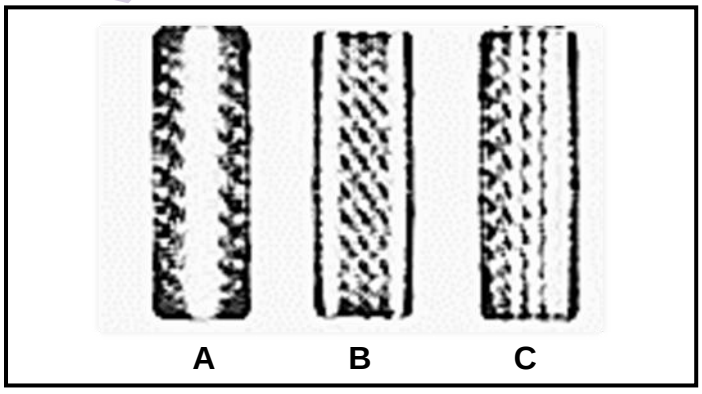


FIGURE 3 (3)

3.5 Describe any TWO facts that you should consider when you replace the battery of a vehicle with a new one. (2)

3.6 Name any THREE vehicle battery parts that are found on the outside of a battery. (3)

3.7 How many volts does each of the cells contribute towards the output voltage of a fully charged battery? (1)

3.8 What is the specific gravity reading of a battery cell for the following conditions?

3.8.1 A half-charged cell

3.8.2 A dead cell

3.8.3 A fully charged cell

(3 × 1) (3)

3.9 Name any FOUR parts of a standard headlight assembly. (4)

3.10 What does the abbreviation LED represent? (1)

[25]

QUESTION 4

4.1 Give TWO reasons why the job card is an important document. (2)

4.2 Name THREE places fluid leaks can occur and for which you need to check during an under body inspection. (3)

4.3 Name TWO workshop equipment beside hand tools that you would need to perform a service on a vehicle. (2)

4.4 Name THREE types of oil the motor mechanic would need to replace or top-up when servicing a vehicle. (3)

4.5 It is important to check the operation of the vehicle's lights during a service.

Name FOUR lights that you should check in the front of the vehicle. (4)

4.6 Name THREE types of waste materials that you would need to dispose of when you restore the work area after the service is completed. (3)

4.7 Give THREE reasons why a vehicle is road tested after it was serviced. (3)

[20]

TOTAL: 100

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